

Putting Everything Together

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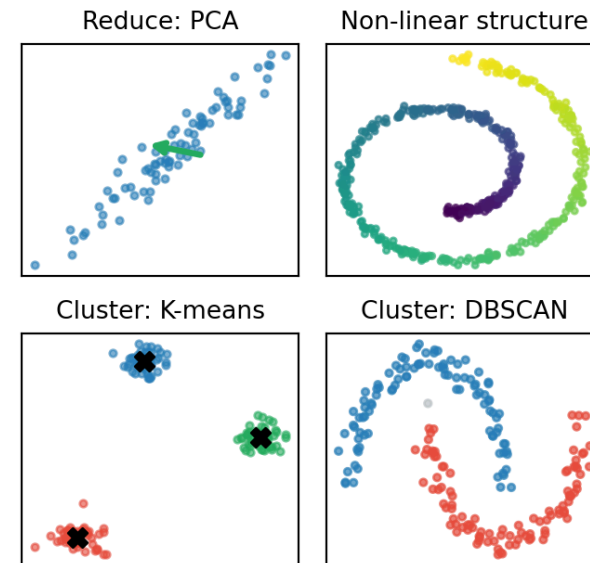
Padma Venkatraman

Building the pipeline

Everything we built collapses into a single workflow:

Scale → **Reduce** → **Cluster** → **Validate** ↻

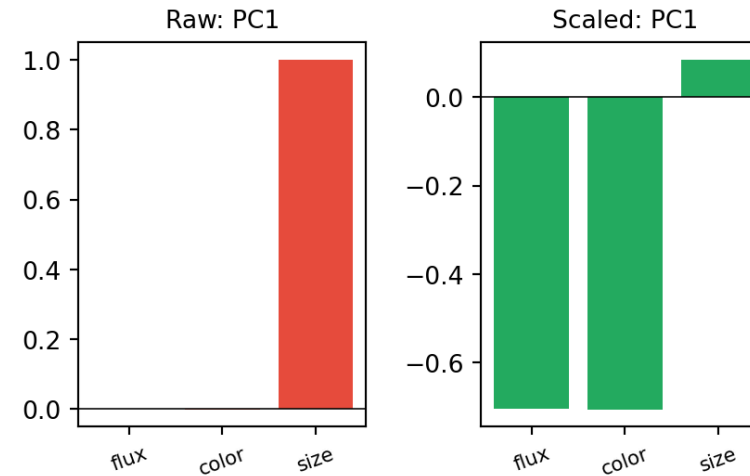
- **Scale** — put features on a common footing
- **Reduce** — strip noisy dimensions (PCA / UMAP)
- **Cluster** — find structure (K-means / GMM / DBSCAN)
- **Validate** — is the structure *real*, and is it the one you care about?



Step 1 — Scale

A feature measured in big numbers gets a big variance — and **hijacks** PCA and clustering, regardless of whether it carries any signal.

- The fix: **standardise** each feature to mean 0, std 1 (the z-score)
- Then no feature dominates by units alone
- *When not to*: if a feature's raw scale is genuinely meaningful

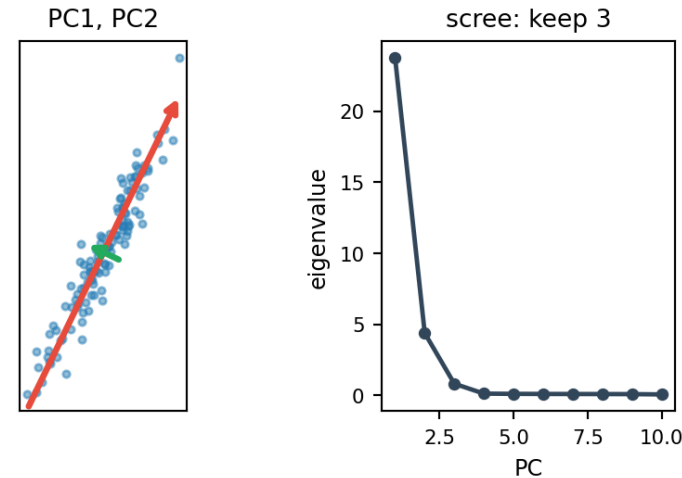


PC1 before scaling : large because of **units**. After scaling you can see the “true” structure.

Step 2 — Reduce

High-dimensional data can have many noise dimensions; Everything can look equally far apart, reduce dimensionality first.

- **PCA** — linear; keeps directions of **global variance**; distances meaningful; new points project straight in
- **UMAP** — non-linear; preserves **local neighbourhoods**; for structure PCA can't unfold
- How many PCs? The **scree elbow** / cumulative variance

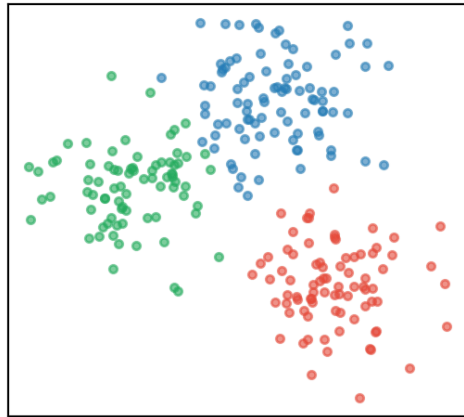


PCA finds the variance axes (left); the scree **elbow** says how many to keep (right).

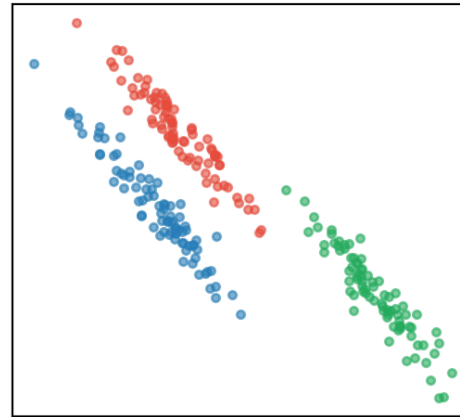
Step 3 — Cluster

Find the appropriate tool to group the points. Each tool has it's own assumptions, which ones are valid for your dataset?

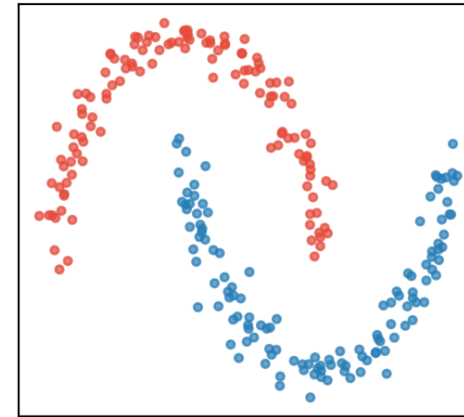
K-means — round, sized-alike, you know K



GMM — elliptical, tilted, soft labels



DBSCAN — any shape + noise, no K

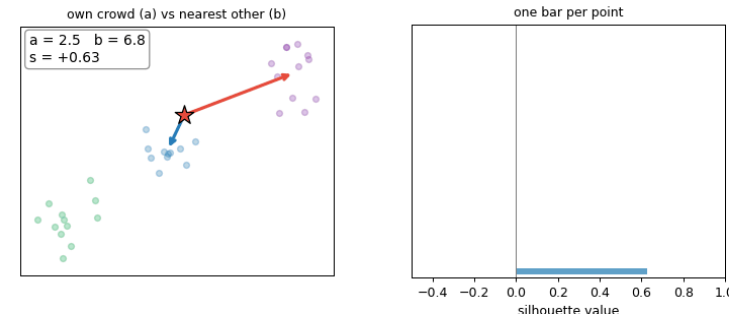


Always **visualise**, and try **more than one** — agreement builds trust, disagreement is a signal.

Step 4 — Validate

Any algorithm returns clusters — even on pure noise. So: are they *real*?

- **Internal** (no labels): the **silhouette**
$$s = \frac{b-a}{\max(a,b)}$$
 — cohesion vs separation,
on $[-1, 1]$
- **External** (when you have labels):
purity, completeness
- The trap: a healthy silhouette scores
the **geometry**, not whether you found
the science class — *necessary, not
sufficient*



Loop!

If the results are bad, try again. Change assumptions, algorithms, parameters.

What you know about the data	Reach for
Round, similar-sized clusters; you know K	K-means
Elliptical / tilted clusters; you know K	GMM
Arbitrary shapes, or outliers to flag	DBSCAN
High-dimensional	reduce first (PCA), then the above
Nothing at all	K-means + elbow, <i>visualise</i> , iterate

A low silhouette or a clustering that looks wrong → re-scale, re-reduce, change algorithm, change K .

Takeaways

- **Scale first** — distance methods are only as fair as your units
- **Reduce** — kill noise dimensions before clustering; PCA for variance, UMAP for local shape
- **Cluster** — match the tool to the data's shape; there is no universally best one
- **Validate** — internal scores check geometry; check against known science whenever you can
- The whole thing is a **loop** you iterate, not a line you walk once